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MATH2056 - LINEAR ALGEBRA

MAKEUP FINAL EXAM =)

"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

QUESTIONS =)

① Consider the matrix $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$. Some of the elimination matrices are given below:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

a) (5p.) Find the unknown entries α, γ, β, k and m = ?

b) (5p.) Find the eliminated upper triangle matrix U = ?

c) (10p.) Determine the remaining elimination matrices.

② A geometry in 't' dimensional space is given by $x = \gamma y$ expression, where 'x' and 'y' represents the axis. The intersection with another geometry gives a volume.

a) (10p.) Give the minimum dimension of the space with your valuable comments.

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b) (10p.) Propose an expression for the unknown geometry to have a volume in intersection.

c) (15p.) Proposed a geometry $x+4y-z=6$. Find all vectors in the intersection. Do these vectors form a vector space? The intersection has how many dimensions inside how many dimensional space?

(3) Suppose u is a unit vector in R^n. This problem is about the n by n symmetric matrix $H = I - 2uu^T$:

a) (5p.) Show directly that $H^2 = I$

b) (5p.) Show that H is an orthonormal matrix.

c) (5p.) One eigen vector of H is u itself. Find the corresponding eigenvalue.

d) (5.p) Is H diagonalizable? Why or why not?

(4) Two tanks are connected, first tank has $v(0)=10$ liters liquid and the second tank has $w(0)=40$ liters of water. The liquid diffusion between the tanks are proportional to the difference $v-w$.

a) (10p.) Which one of the following differential equation set expresses the process? (v_L: the liquid diffused to the second tank, w_L: the water diffused to the second tank.)

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_L \\ w_L \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

b) (10p.) Determine peasentages of the mixture in tank1 at time $t=1$sec.

c) (10p.) Find how much liters of liquids are exist in each tank in steady state?

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SOLUTIONS -)

(2) k dim. space $x = 5y$ $x, y \rightarrow$ axis

intersection with another geometry → a volume = 3 dim. space

a) "k" ve diğer geometri en az 3 boyutlu olmalı, çünkü kesişimlerinin hacim bildirmesi için ortak 3 axis'e sahip olmaları gerekir. En azından ortak olan 3 axislerinden dolayı ikisinin de 3 boyutlu olması gerekir. //

b) Bilinmeyen geometry en az 3 boyutlu olmalı ve bu 3 boyutta x, y ve $x = 5y$'nin içinde bulunduğu uzaydan ortak bir yönü daha olmalıdır.

örneğin → $A = \begin{bmatrix} 1 & -5 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0$

diğer geometri $= B = \begin{bmatrix} a & b & 0 & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0$

$x + y - 3z = 0$ gibi //

c) $x - 5y = 0$
$x + 4y - z = 6$ $\Rightarrow R_3$

$$\begin{bmatrix} 1 & -5 & 0 \\ 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

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$$\left[\begin{array}{ccc|c} 1 & -7 & 0 & 0 \\ 1 & 4 & -1 & 6 \end{array}\right] \xrightarrow{R_2-R_1=R_2} \left[\begin{array}{ccc|c} 1 & -7 & 0 & 0 \\ 0 & 9 & -1 & 6 \end{array}\right] \xrightarrow{R_2/9=R_2}$$

$$\left[\begin{array}{ccc|c} \textcircled{1} & -7 & 0 & 0 \\ 0 & \textcircled{1} & -1/9 & 2/3 \end{array}\right] \xrightarrow{R_1+5R_2=R_1} \left[\begin{array}{ccc|c} \textcircled{1} & 0 & -7/9 & 10/3 \\ 0 & \textcircled{1} & -1/9 & 2/3 \end{array}\right] \Rightarrow \begin{array}{c} 2 \text{ pivot} \\ \| \\ 2 \dim C(A) \end{array}$$

$\Rightarrow$ There is infinite solution

$$\left[\begin{array}{ccc} 1 & 0 & -7/9 \\ 0 & 1 & -1/9 \end{array}\right] \begin{bmatrix} x \\ y \\ z \end{bmatrix} \longrightarrow \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \gamma \begin{bmatrix} -7/9 \\ -1/9 \end{bmatrix}$$

are vectors in the intersection.

$\Rightarrow$ The intersection has 2 dim $C(A)$ inside R_3. //

(4)

1.tank
$v(0)=10$
$+(v-w)b$

2.tank
$w(0)=40$
$-(v-w)b$

$v_k = +(v-w)b$
$w_k = w-(v-w)b$

a) $v_{k+1} = (v-w)b \rightarrow vb-wb \longrightarrow bv(t)-bw(t)$

$w_{k+1} = w_k-(v-w)b \longrightarrow -vb+(1+b)w \longrightarrow -bv(t)+(1+b)w(t)$

$$\frac{dv}{dt} = vb-wb \qquad \frac{dw}{dt} = -bv+(1+b)w \qquad \begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} b & -b \\ -b & 1+b \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$$

$$\Rightarrow \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} b & -b \\ -b & 1+b \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

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when we take $b=1 \rightarrow \begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

En yakın cevap $= \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$ //

b) I take this differential equation set: $\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$

$A = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}$ $\quad |\lambda I - A| = 0$

$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$ $\quad \lambda I - A = \begin{bmatrix} \lambda-2 & 2 \\ 1 & \lambda-1 \end{bmatrix} = 0$

$(\lambda-2)(\lambda-1) - 2 = 0$

$\lambda^2 - 3\lambda = \lambda(\lambda-3) = 0$ $\quad \lambda_1 = 0 \quad \lambda_2 = 3$

$\lambda_1 = 0 \rightarrow -A + \lambda I = \begin{bmatrix} -2 & 2 \\ 1 & -1 \end{bmatrix}, x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2 = 3 \rightarrow \lambda I - A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad x_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} v_n \\ w_n \end{bmatrix} = c_1 \cdot x_1 \cdot \lambda_1^n + c_2 \cdot x_2 \cdot \lambda_2^n = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} 0^n + c_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} 3^n$

$n=0 \rightarrow \begin{bmatrix} v(0) \\ w(0) \end{bmatrix} = \begin{bmatrix} 10 \\ 40 \end{bmatrix} \rightarrow \begin{bmatrix} 10 \\ 40 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} 0^0 + c_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot 3^0$

$\begin{bmatrix} 10 \\ 40 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ $\quad c_1 + 2c_2 = 0$, $c_1 - c_2 = 40$ $\Rightarrow$ $c_2 = -10$, $c_1 = 30$

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$$\Rightarrow \begin{bmatrix} v_k \\ w_k \end{bmatrix} = 30\begin{bmatrix} 1 \\ 1 \end{bmatrix} 0^n - 10\begin{bmatrix} 2 \\ -1 \end{bmatrix} 3^n$$

$$n \neq 0 \Rightarrow \begin{bmatrix} v_k \\ w_k \end{bmatrix} = -10\begin{bmatrix} 2 \\ -1 \end{bmatrix} 3^n$$

$t = 1\ sec \Rightarrow \frac{1}{60} = n \qquad v_k\left(\frac{1}{60}\right) = +20.37 \qquad \frac{v_k}{w_k} = \frac{20.37}{10.19} = 2 //$

$w_k\left(\frac{1}{60}\right) = 10.19$

c) $n \to \infty \Rightarrow 3^n \to \infty$

$$\begin{bmatrix} v_k \\ w_k \end{bmatrix} = \begin{bmatrix} \infty \\ \infty \end{bmatrix} //$$